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PHYSICS

1) Two spheres of radius a and b respectively are charged and joined by a wire. The ratio of electric field of the spheres is:-

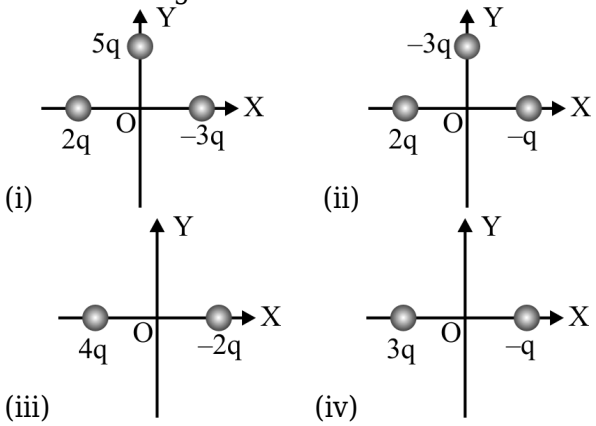
(1) a/b

(2) b/a

(3) a^2/b^2

(4) b^2/a^2

2) In the following four situations charged particles are at equal distance from the origin. Arrange them the magnitude of the net electric field at origin greatest first :-



(1) (i) > (ii) > (iii) > (iv)

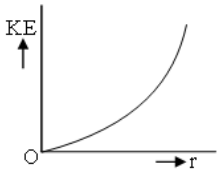
(2) (ii) > (i) > (iii) > (iv)

(3) (i) > (iii) > (ii) > (iv)

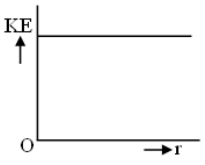
(4) (iv) > (iii) > (ii) > (i)

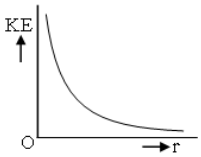
3) An infinitely long positively charged straight thread has a linear charge density λ Cm^{-1}. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :

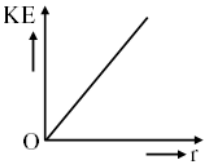
(1)

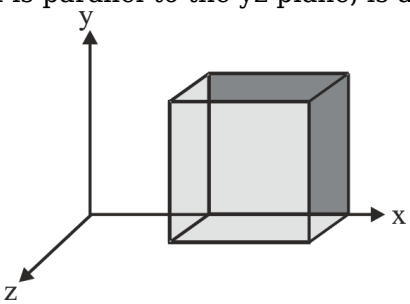


(2)



(3) 

(4) 

4) A cube of side a is placed such that the nearest face, which is parallel to the yz plane, is at a distance a from the origin. The electric field components are 

$E_x = \alpha x^{1/2}$, $E_y = E_z = 0$.
The flux ϕ_E through the cube is :-

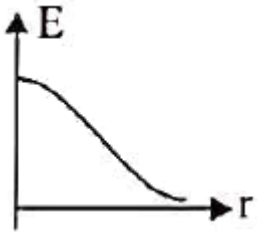
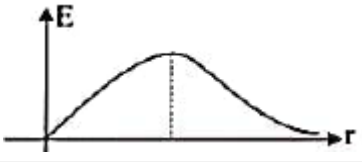
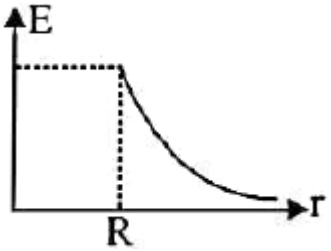
(1) $2\sqrt{2}\alpha a^{5/2}$

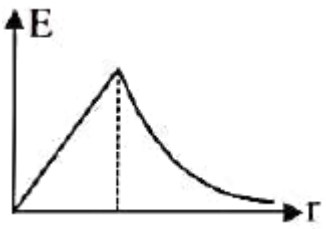
(2) $-\alpha a^{5/2}$

(3) $(\sqrt{2}-1)\alpha a^{5/2}$

(4) zero

5) Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in column I

	Column-I		Column-II
(A)	Ring along its axis	(P)	
(B)	Uniformly charged solid sphere	(Q)	
(C)	Uniformly charged spherical shell	(R)	

(D)	Combination of charge +Q and – Q at the perpendicular bisector	(S)	

(1) (A) - (Q), (B)-(S), (C)-(R), (D)-(P)

(2) (A) - (Q), (B)-(S), (C)-(P), (D)-(S)

(3) (A) - (R), (B)-(P), (C)-(Q), (D)-(R)

(4) (A) - (P), (B)-(Q), (C)-(S), (D)-(Q)

6)

The potential function of electrostatic field is given by $V = 2x^2 - 3xy^2z + 4yz$. Then electric field at point (0, 1, 1)m will be

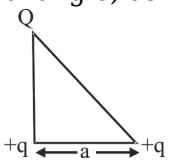
(1) $3\hat{i} - 4\hat{j} + 4\hat{k}$

(2) $2\hat{i} - 4\hat{k}$

(3) $3\hat{i} - 4\hat{j} - 4\hat{k}$

(4) $-3\hat{i} + 4\hat{j} + 4\hat{k}$

7) Three charges Q, +q and +q are placed at the vertices of a right-angled triangle (isosceles triangle) as shown. The net electrostatic energy of the configuration is zero, if Q is equal to :



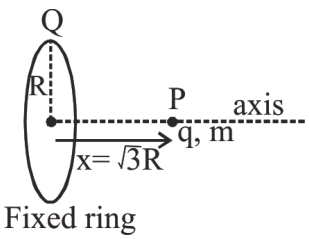
(1) $\frac{-q}{1+\sqrt{2}}$

(2) $\frac{-2q}{2+\sqrt{2}}$

(3) –2q

(4) +q

8) A point charge 'q' of mass is located at the axis of a ring at point P($x = \sqrt{3}R$) oriented along x-axis. When the point charge q is given negligible velocity. The speed of charge particle when it reaches to infinity is :-



Fixed ring

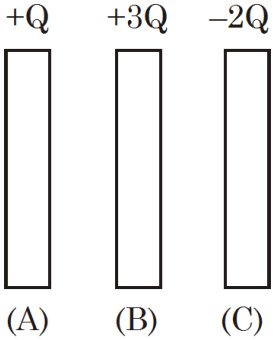
(1) $v = \sqrt{\frac{KQq}{mR}}$

(2) $v = \sqrt{\frac{2KQq}{mR}}$

(3) $v = \sqrt{\frac{KQq}{2mR}}$

(4) None of these

9) The ratio of charge on left surface of plate A to the charge on the right surface of plate B is :-



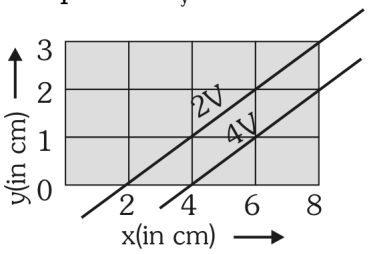
(1) 0

(2) $\frac{1}{3}$

(3) $\frac{2}{3}$

(4) $-\frac{1}{3}$

10) Figure shows two equipotential lines in xy-plane for an electric field. The x-component E_x and y-component E_y of the field in the electric space between these equipotential lines are, respectively -



(1) + 100 V/m, –200 V/m

(2) –100 V/m, + 200 V/m

(3) + 200 V/m, 100 V/m

(4) –200 V/m, –100 V/m

11) An electric field $\vec{E} = 20\hat{i} + 30\hat{j}\ \frac{N}{C}$ exist in the space. The potential at origin is taken to be 150 V. Then the potential at (2m, 2m) is :-

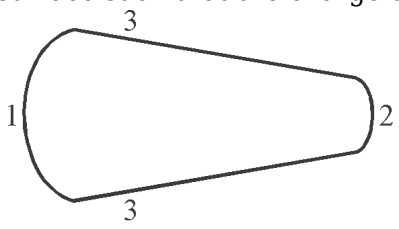
(1) 100 V

(2) Zero

(3) 50 V

(4) 200 V

12) Consider a non-spherical conductor shown in the figure which is given a certain amount of positive charge. The charge distributes itself on the surface such that the charge densities are σ_1, σ_2 and σ_3 at the regions 1, 2 and 3 respectively. Then :-



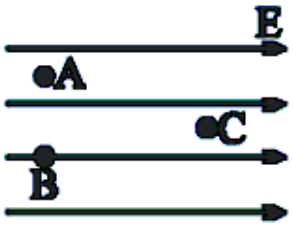
(1) $\sigma_1 > \sigma_2 > \sigma_3$

(2) $\sigma_2 > \sigma_3 > \sigma_1$

(3) $\sigma_3 > \sigma_1 > \sigma_2$

(4) $\sigma_2 > \sigma_1 > \sigma_3$

13) Suppose a region of space has a uniform electric field, directed towards the right, as shown below. Which statement is true?



(1) The potential at all three locations is the same

(2) The potential at points A and B are equal and the potential at point C is higher than the potential at point A

(3) The potential at points A and B are equal, and the potential at point C is lower than the potential at point A

(4) The potential at point A is the highest, the potential at point B is the second highest, and the potential at point C is the lowest.

14) There is a uniform electric field in X-direction. If the work done by external agent in moving a charge of 0.2 C through a distance of 2 metre slowly along the line making an angle of 60° with X-direction is 4 joule, then the magnitude of E is :

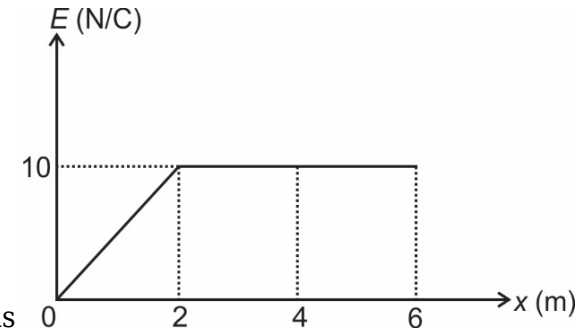
(1) $\sqrt{3}$ N/C

(2) 4 N/C

(3) 5 N/C

(4) 20 N/C

15) Figure shows the variation of electric field intensity E versus distance x. The magnitude of

potential difference between the points x = 0 to x = 5 m is 

(1) 30 volt

(2) 60 volt

(3) 40 volt

(4) 50 volt

16) A $500\mu F$ capacitor is charged at a steady rate of 100μC/ second. The potential difference across the capacitor will be 10V after an interval of :-

(1) 5 sec

(2) 25 sec

(3) 20 sec

(4) 50 sec

17) The radius of the circular plates of a parallel plate capacitor is 'R'. Air is dielectric medium between the plates. If the capacitance of the capacitor is equal to the capacitance of sphere of radius 'R', then the distance between the plates is :

(1) $\frac{R}{4}$

(2) $\frac{R}{2}$

(3) R

(4) 2R

18) A 40 mF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is :-

(1) 45 kW

(2) 90 MW

(3) 180 kW

(4) 360 kW

19) A parallel plate capacitor of capacitance C is having charge q_0 on each plate. Now it is connected to a battery of emf V then which option is correct ?
(i) Charge supplies by battery would be CV
(ii) The outer surface of plate have equal charge q_0
(iii) The facing surface have equal and opposite charge of magnitude CV

(1) i, ii

(2) i, iii

(3) ii, iii

(4) i, ii, iii

20) The plates of a parallel plate capacitor are pulled apart with a velocity v. If at any instant their mutual distance of separation is x then magnitude of rate of change of capacitance with respect to time varies as :

(1) $\frac{1}{x}$

(2) $\frac{1}{x^2}$

(3) x^2

(4) x

21)

If the plates of a parallel plate capacitor are not equal in area, then:

(1) quantity of charge on the plates will be same but nature of charge will be different

(2) quantity of charge on the plates as well as nature of charge will be different

(3) quantity of charge on the plates will be different but nature of charge will be same

(4) quantity of charge as well as nature of charge will be same

22) A battery is used to charged a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will doubled the :

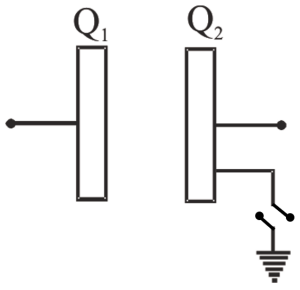
(1) capacitance

(2) surface charge density on each plate

(3) stored energy

(4) electric field between the two places

23) The two plates of a parallel plate capacitors charges Q_1 and Q_2 are given.The capacity of the capacitor is C. When the switch is closed, mark the correct statement(s).Assuming both Q_1,Q_2 to be



positive.

(1) The charge flown through switch is zero.

(2) The charge flown through switch is $\frac{Q_1 + Q_2}{2}$

(3) Potential difference across the capacitor plates is $\frac{Q_1}{C}$.

The charge of the capacitor is

(4) $\frac{Q_1}{2}$

CHEMISTRY

1) Which of the following statements are correct?
A. The process of adding an electron to a neutral gaseous atom is always exothermic.
B. The process of removing an electron from an isolated gaseous atom is always endothermic.
C. The 1st ionization energy of boron is less than that of beryllium.
D. The electronegativity of C is 2.5 in CH_4 and CCl_4
E. Li is the most electropositive among elements of group I.
Choose the correct answer from the options given below:

(1) A, C and D Only

(2) B and C Only

(3) B and D Only

(4) B, C and E Only

2) Given below are two statements:
Statement (1): The metallic radius of Al is less than that of Ga.
Statement (II): The ionic radius of Al^{3+} is less than that of Ga^{3+}
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are correct

(3) Statement I is incorrect but Statement Il is correct

(4) Both Statement I and Statement II are incorrect

3) An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}. 'E' reacts with elements A, B, C and D with electron gain enthalpy values of –328, –349, –325 and –295 kJ mol^{-1}, respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is:

(1) EB > EA > EC > ED

(2) EA > EB > EC > ED

(3) ED > EC > EA > EB

(4) ED > EC > EB > EA

4) Which one of the following order represents the correct sequence of the increasing basic nature of the given oxides ?

(1) $MgO < K_2O < Al_2O_3 < Na_2O$

(2) $Al_2O_3 < MgO < Na_2O < K_2O$

(3) $Na_2O < K_2O < MgO < Al_2O_3$

(4) $K_2O < Na_2O < Al_2O_3 < MgO$

5)

The electronic configuration of four elements are :
(I) $[Kr]5s^1$ (II) $[Rn]5f^{14}6d^17s^2$
(III) $[Ar]3d^{10}4s^24p^5$ (IV) $[Ar]3d^64s^2$
Consider the following statements :
(i) I shows variable oxidation state
(ii) II is a d-block element
(iii) The compound formed between I and III is covalent
(iv) IV shows single oxidation state

(1) FTFF
(2) FTFT
(3) FFTF
(4) FFFF

6)

The variation of the boiling points of the hydrogen halides is in the order HF > HI > HBr > HCl.

What explains the higher boiling point of hydrogen fluoride ?
(1) The bond energy of HF molecules is greater than in other hydrogen halides
(2) The effect of nuclear shielding is much reduced in fluorine which polarises the HF molecule
(3) The electronegativity of fluorine is much higher than for other elements in the group.
(4) There is strong hydrogen bonding between HF molecules

7) Consider the following electron addition process
(i) $O \xrightarrow{\Delta H_1} O^- \xrightarrow{\Delta H_2} O^{2-}$
(ii) $S \xrightarrow{\Delta H_3} S^- \xrightarrow{\Delta H_4} S^{2-}$
(iii) $P \xrightarrow{\Delta H_5} P^- \xrightarrow{\Delta H_6} P^{2-}$
The which of the following statement is incorrect regarding given information:

(1) ΔH_5 is less negative than ΔH_3 and ΔH_1
(2) ΔH_1 is less negative than ΔH_3
(3) ΔH_2 is more positive than ΔH_4
(4) none of these

8) Which the following interaction form nonbonding molecular orbital when z-axis is the bonding axis?

(1) $d_{yz} + d_{z^2}$
(2) $d_{yz} + d_{xy}$
(3) $d_{x^2-y^2} + d_{xy}$
(4) All form N.B.M.O.

9) Select the incorrect statement:

(1) If the orbitals differ largely in energy, the cost of hybridization energy becomes large

(2) Phosphorus has vacant 3d orbitals

(3) The d-orbital participation generally requires to bond with highly electronegative elements

(4) PH_5 does not exist but PCl_5, PF_5 exist through the formation of sp^3d^2 hybridization

10) Match the following:

	Column-I		**Column-II (Dominating force)**
(a)	Xenon-H_2O	(p)	Dipole-dipole interaction
(b)	Ethanol-H_2O mixture	(q)	London forces
(c)	Liquid He	(r)	Ion-dipole interaction
(d)	Liquid HCl	(s)	Dipole-induced dipole interaction

(1) a-s; b-q; c-p; d-p

(2) a-s; b-p; c-q; d-p

(3) a-p; b-q; c-p; d-r

(4) a-s; b-r; c-q; d-p

11) Match the following:

Column-I		**Column-II**	
(a)	NH_2^-	(p)	$\mu = 0$
(b)	NH_4^+	(q)	Planar
(c)	I_3^-	(r)	$d_{x^2-y^2}$ is involved in hybridisation
(d)	PCl_6^-	(s)	d_{z^2} is involved in hybridisation
(e)	PCl_2^+	(t)	Bent structure

(1) a → q, p; b → p; c → p, s; d → r, s ; e → t

(2) a → q; b → p, q; c → p, s, r; d → r, s ; e → t, r

(3) a → q, t; b → p; c → p, q, s; d → p, r, s ; e → q, t

(4) a → q; b → p; c → p, q; d → p, r, s ; e → q, t

12) Consider the following statements and select the correct set of codes using T(True) and F(False) in the given sequence.

(I) PI_5(s) does not exist due to steric crowding.

(II) All possible angle in BF_2Cl are 120°.

(III) In $N(SiH_3)_3$, lone pair of N is present in sp^3 hybrid orbital.

(IV) d orbital used in hybridization of P in PBr_5(s) is d_{z^2}

(1) TFTT

(2) FTTT

(3) FFFF

(4) TFTF

13) Among SO_2, NF_3, NH_3, XeF_2, ClF_3 and SF_4, the hybridization of the molecule with nonzero dipole moment and highest number of lone-pairs of electrons on the central atom is

(1) sp^3

(2) dsp^2

(3) sp^3d^2

(4) sp^3d

14) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3.
Reason (R): In NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N—H bonds. F is the most electronegative element.
In the light of the above statements, choose the correct answer from the options given below:

(1) (A) is false but (R) is true

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

15) According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are

(1) 1,3 and 3

(2) 2, 3 and 3

(3) 1, 2 and 3

(4) 1,3 and 2

16) Consider the following table

Total number of electron pairs (l.p. + σ-bond)	**Total number of lone pairs**	**Shape**
5	p	Linear
q	1	See-saw
4	r	Bent shape
s	2	Square planar
5	t	Bent "T" shape

Then calculate value of "p + q + r – s – t'.

(1) 2

(2) 10

(3) 0

(4) 5

17)

Which of the following statement is correct ?

(1) Ortho nitrophenol is less volatile than para nitrophenol.
(2) Ortho nitrophenol has more boiling point than para nitrophenol.
(3) Para nitrophenol has more boiling point than ortho nitrophenol.
(4) Ortho and para nitrophenol has same volatility

18) On applying pressure to the equilibrium,
ice ⇌ water
Which phenomenon will happen?

(1) More ice will be formed
(2) More water will be formed
(3) Equilibrium will not be disturbed
(4) Water will evaporate

19) For $H_2 + I_2 \rightleftharpoons 2HI$, at equilibrium some I_2 is added.
What happens to the equilibrium?

(1) It is shifted to the right
(2) It gets shifted to the left
(3) It remains unchanged
(4) None of the above

20) We know that the relationship between K_C and K_P is $K_P = K_C (RT)^{\Delta n}$. What would be the value of Δn for the reaction $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$?

(1) 1
(2) 0.5
(3) 1.5
(4) 2

21) PCl_5, PCl_3 and Cl_2 are at equilibrium at 500 K in a closed container and their concentrations are 0.8×10^{-3} mol L^{-1}, 1.2×10^{-3} mol L^{-1} and 1.2×10^{-3} mol L^{-1} respectively. The value of K_C for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ will be:

(1) 1.8×10^{3} mol L^{-1}
(2) 1.8×10^{-3} mol L^{-1}
(3) 1.8×10^{-3} L mol^{-1}
(4) 0.55×10^{4} mol L^{-1}

22) Densities of diamond and graphite are 3.5 and 2.3 g/mL respectively. Increase of pressure on the equilibrium $C_{diamond} \rightleftharpoons C_{graphite}$:

(1) Favours backward reaction

(2) Favours forward reaction

(3) Have no effect

(4) Increases the reaction rate

BIOLOGY

1) Which bacteria oxidise various inorganic substances and uses the released energy for their ATP production ?

(1) Symbiotic bacteria

(2) Photosynthetic bacteria

(3) Chemosynthetic bacteria

(4) Heterotrophic bacteria

2) Members of which kingdom play a great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur?

(1) Monera

(2) Algae

(3) BGA

(4) Bryophyta

3) The basis of modern taxonomic studies are :-
(a) External and internal structure
(b) Only external structure
(c) Development process
(d) Ecological informations

(1) a and b only

(2) a, b and c only

(3) a, c and d only

(4) All a, b, c and d

4) The earliest classifications were based on :-

(1) Uses of organisms

(2) Internal structure of organism

(3) Their phylogeny

(4) Their gross morphology

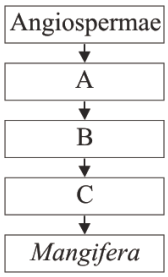
5) Which statement is/are correct.
(a) Biological names are generally in Latin and written in italics.
(b) Each taxon in fact, represents a unit of classification. (c) *Datura* and *Petunia* belong to solanaceae.

(1) 'a' only

(2) a,b,c

(3) *a,c*

(4) 'c' only



6) Choose the correct option for the given taxonomical hierarchy :-

(1) A-Monocotyledonae, B-Sapindales, C-Anacardiaceae

(2) A-Dicotyledonae, B-Anacardiaceae, C-Sapindales

(3) A-Dicotyledonae, B-Sapindales, C-Anacardiaceae

(4) A-Monocotyledonae, B-Anacardiaceae, C-Sapindales

7)

Arrange the following taxon (a, b, c, d) according to their taxonomic ranks in descending manner :-
(a) *Panthera* (b) Mammalia

(c) *Felidae* (d) Carnivora

(1) b→c→a→d

(2) a→b→c→d

(3) d→c→b→a

(4) b→d→c→a

8) The bacterial cell is very simple in ___A___ but very complex in ___B___.
Select the option in which A and B are correctly identified.

	A	**B**
(1)	Behaviour	Structure
(2)	Reproduction	Structure
(3)	Shape	size
(4)	Structure	Behaviour

(1) 1

(2) 2

(3) 3

(4) 4

9) Match the column-I with column-II and choose **correct** answer :-

	Column-I		Column-II
(P)	*Ciliated protozoans*	(i)	Pseudopodia
(Q)	*Sporozoans*	(ii)	Sleeping sickness
(R)	*Flagellated protozoans*	(iii)	*Paramoecium*
(S)	*Amoeboid protozoans*	(iv)	Malaria parasite

(1) P-iii, Q-iv, R-i, S-ii

(2) P-iii, Q-i, R-iv, S-ii

(3) P-iii, Q-ii, R-iv, S-i

(4) P-iii, Q-iv, R-ii, S-i

10) In which of the following, the genus name, its character and its class are not correctly matched, whereas the remaining three are correctly matched ?

	Genus Name	Character	Class
(1)	*Alternaria*	Sexual stage is absent	Deuteromycetes
(2)	*Aspergillus*	Endogenous ascospores	Ascomycetes
(3)	*Mucor*	Aseptate mycelium	Phycomycetes
(4)	*Albugo*	septate and coenocytic mycelium	Ascomycetes

(1) 1

(2) 2

(3) 3

(4) 4

11) If a bacteria divides once in every 3 minutes and fill the cup in 57 minutes. How much time it will take to fill 1/4 of the cup ?

(1) 54 minutes

(2) 51 minutes

(3) 3 minutes

(4) 14 minutes

12) Red tide is caused by :-

(1) *Gracilaria*

(2) *Chlamydomonas*

(3) *Gonyaulax*

(4) *Noctiluca*

13) Choose the incorrect pair.

(1) Gram positive bacteria - Take up the gram stain
(2) Cell wall - Prevents bacterium from collapsing
(3) Capsule - Thick and tough glycocalyx in bacteria
(4) Pili - Locomotory structure in bacteria

14) How many of the following belongs to sac fungi? *Penicillium, Aspergillus, Claviceps, Morels, Neurospora, Yeast, Agaricus, Ustilago* and Mushroom.

(1) Five
(2) Six
(3) Seven
(4) Eight

15) **Assertion:** Cyanobacteria are photosynthetic archaebacteria, which is having all prokaryotic structures.
Reason: They are only green in colour because of the presence of chloroplast.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

16)

Find out the correct match from the following table :

	Column-I	**Column-II**	**Column-III**
(i)	Mesosomes	Oxidative enzyme	Cell respiration
(ii)	Capsule	Loose sheath	Protein synthesis
(iii)	Flagella	Flagellin protein	Help in genetic recombination

(1) (i) and (ii)
(2) (ii) and (iii)
(3) only (iii)
(4) only (i)

17) **Statement I :** Kingdom Protista under five kingdom classification has brought together *Chlamydomonas*, *Chlorella* with *Paramoecium* and Amoeba. It has put together organisms which, in earlier classifications, were placed in different kingdoms.
Statement II : This happened because the criteria for classification changed over time.

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

18) Presense of Pyrenoids is character of algae, a pyrenoid consists of ?

(1) Starch besides protein

(2) Protein besides starch

(3) Core of fatty acid surrounded by protein

(4) Glycogen and protein

19) Match List-I with List-II.

	List-I (Type of Sexual Reproduction)		**List-II (Examples)**
(a)	Filamentous Algae of Chlorophyceae	(i)	*Kelps*
(b)	Filamentous Algae of Phaecophyceae	(ii)	*Volvox*
(c)	profusely branched forms of Phaeophyceae	(iii)	*Ectocarpous*
(d)	Colonial Algae of Chlorophyceae	(iv)	*Ulothrix*

Choose the correct answer from the options given below:

(1) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(3) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

(4) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)

20) Which fungi cause rust and smut disease respectively:-

(1) *Penicillium* and *Ustilago*

(2) *Puccinia* and *Ustilago*

(3) *Puccinia* and *Erysiphe*

(4) *Ustilago* and *Puccinia*

21) The protistans having protein rich layer which makes their body flexible are :-

(1) Slime moulds

(2) Chrysophytes

(3) Dinoflagellates

(4) Euglenoids



22) Find correct option regarding this diagram: (a) Colour of this organism

appear as green only.
(b) Releases toxins that kills fishes and other marine animal
(c) Two flagella differ in their position
(d) Cell wall is made of peptidoglycan.

(1) a, b
(2) b and c
(3) a, b and c
(4) b, c and d

23) **Assertion :-** *Neurosphora* is used extensively in biochemical and genetic work.
Reason :- *Neurospora* grow fast, easily cultivate and has short life span.

(1) Both Assertion and Reason are true and Reason is correct explanation of the Assertion.
(2) Both Assertion and Reason are true but R is not correct explanation of the Assertion.
(3) Assertion is true and Reason is false.
(4) Assertion and Reason both are false

24) **Assertion :** Deuteromycetes are also known as imperfect fungi.
Reason : Only the asexual or vegetative phases of deuteromycetes are known.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

25) Read the following statements regarding methanogens and select the correct option :
(i) They are included in the group Archaebacteria
(ii) They are responsible for the production of biogas in gobar gas plants
(iii) They live in hot sulphur springs
(iv) They are anaerobic

(1) Statement (i) and (ii) are correct
(2) Statement (i), (ii) and (iv) are correct
(3) Statement (i), (ii) and (iii) are correct
(4) Statement, (ii), (iii) and (iv) are correct

26) In fern, prothallus develops :-

(1) From gamete
(2) From zygote
(3) By secondary protonema
(4) From spore

27) Match column A with column B -

Column A	Column B

a	Mycorrhiza	i	Pollengrain
b	Coralloid roots	ii	*Pinus*
c	Male gametophyte	iii	*Cycas*
d	*Selaginella*	iv	*Sequoia*
e	Giant redwood	v	Lycopsida

(1) a-(i), b-(ii), c - (iii), d- (iv), e - (v)

(2) a-(v), b-(iv), c - (iii), d- (ii), e - (i)

(3) a-(ii), b-(iii), c - (i), d- (v), e - (iv)

(4) a-(iv), b-(iii), c - (v), d- (i), e - (ii)

28) Mosses are attached to soil through ___(A)___ and ___(B)___ rhizoids.

(1) A-Branched, B-unicellular

(2) A-Unicellular, B-branched

(3) A-Multicellular, B-branched

(4) A-Multicellular, B-unbranched

29) Which of the following is not an edible algae ?

(1) *Porphyra*

(2) *Laminaria*

(3) *Sargassum*

(4) *Fucus*

30) Which of the following not belongs to similar taxonomic group ?

(1) *Polysiphonia*

(2) *Porphyra*

(3) *Sargassum*

(4) *Gracilaria*

31) Match the column :-

Column-I		**Column-II**	
A	*Cycas*	i	Heterosporous fern
B	*Pinus*	ii	Homosporous fern
C	*Salvinia*	iii	Pinnate leaves
D	*Adiantum*	iv	Needle like leaves

(1) A → iv; B → iii; C → ii; D → i

(2) A → iv; B → iii; C → i; D → ii

(3) A → iii; B → iv; C → i; D → ii

(4) A → iii; B → iv; C → ii; D → i

32) Choose the correct statement regarding gymnosperm -

(1) Seeds are developed before endosperm formation

(2) Ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilization

(3) They show double fertilization

(4) They form haploid endosperm after fertilization

33) In pteridophytes, Prothallus produces :-

(1) Sporangia

(2) Antheridia and archegonia

(3) Root, Stem and leaf

(4) Leafy gametophyte

34) Choose the **incorrect** statement from the following:-

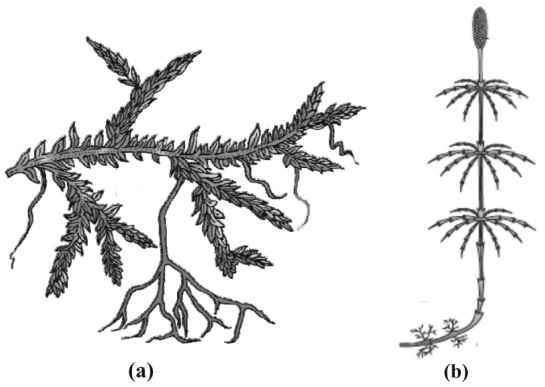
(1) Main plant body of pteridophytes is differentiated into root, stem and leaves.

(2) Large and small types of leaves are present in pteridophytes

(3) Pteridophytes are the first terrestrial plants to possess xylem and phloem.

(4) Most of the pteridophytes produce two kinds of spores, macro and micro spores by meiotic division.

35)



Identify the plants given in the figure (a) and (b).

(1) *Selaginella* and *Pteris*

(2) *Equisetum* and *Salvinia*

(3) *Selaginella* and *Equisetum*

(4) *Salvinia* and *Azolla*

36) Which of the followings is/are not an ecological importance of mosses ?
(A) They form dense mats on soil, thus they reduce the impact of falling rain and prevent soil erosion.
(B) They provide food for herbivore mammals, birds and other animals
(C) They are used as packing material for trans shipment of living materials
(D) Along with lichen they are first to colonise rocks.

(1) A and B

(2) B and C

(3) only C

(4) C and D

37) **Statement–1 :-** Bryophytes are also called amphibians of plant kingdom. **Statement–2 :-** Bryophytes can live in soil but are dependent on water for fertilization.

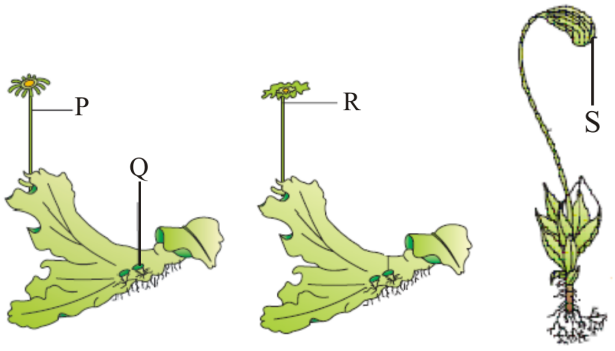
(1) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1.

(2) Statement–1 is false, Statement–2 is true.

(3) Statement–1 is true, Statement–2 is false.

(4) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

38) Observe the following figures and identify the labelled structures P, Q, R, and S:-



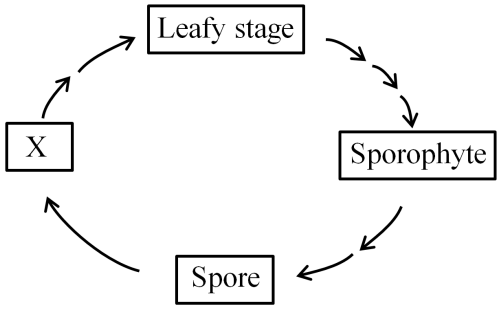
	P	**Q**	**R**	**S**
(1)	Archegonio- phore	Gemma cup	Antheridio- phore	Seta
(2)	Antheridio-phores	Rhizoids	Archegonio-phore	Capsule
(3)	Antheridio-phore	Gemma cup	Archegonio-phore	Capsule
(4)	Archegonio-phore	Gemma cup	Antheridio-phore	Capsule

(1) 1

(2) 2

(3) 3

(4) 4



39) The following flow chart related to life cycle of moss :-
Select the correct option with respect to "X" given in flow chart (life cycle) :-

(1) Selender axis bearing spirally arranged leaves.
(2) "X" have multicellular root
(3) This stage bear the sex organs.
(4) Creeping green, branched and filamentous stage

40) In the list given below, how many members belong to chlorophyceae ?
Porphyra, *Fucus*, *Volvox*, *Chlorella*, *Dictyota*, *Chara*, *Ulothrix*

(1) Three
(2) Four
(3) Two
(4) One

41) Which of the following alga is rich in proteins and used as food supplement by space travellers ?

(1) *Spirulina*
(2) *Chlorella*
(3) *Laminaria*
(4) *Gelidium*

42) **Assertion :** In *Fucus*, sexual reproduction takes place through fusion between one large, non-motile female gamete and a smaller, motile male gamete. **Reason :** *Fucus* show Oogamous type of sexual reproduction.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

43) The algae whose gametes are pyriform and bear two laterally attached flagella :-

(1) Green algae
(2) Red algae
(3) Brown algae
(4) Blue green algae

44)

Arrange the following events in a correct sequence in the life cycle of fern :-
(I) Fertilization (II) Liberation of spores
(III) Prothallus (IV) Embryo

(1) II, III, I, IV
(2) IV, III, II, I
(3) I, II, III, IV
(4) I, IV, III, II

45) Laminarin and mannitol are stored food of :-

(1) Diatoms
(2) Brown algae
(3) Red algae
(4) Dinoflagellates

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	3	1	3	2	1	2	2	3	4	3	4	3	4	1	2	4	2
Q.	21	22	23																	
A.	1	3	3																	

CHEMISTRY

Q.	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43
A.	2	3	1	2	4	4	4	4	4	2	3	3	4	4	1	1	3	2	1	4
Q.	44	45																		
A.	2	3																		

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	3	1	2	3	4	4	4	4	2	3	4	2	4	4	4	2	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	1	1	2	4	3	3	4	3	3	2	2	4	3	3	4	4	4	2
Q.	86	87	88	89	90															
A.	2	1	3	1	2															

SOLUTIONS

PHYSICS

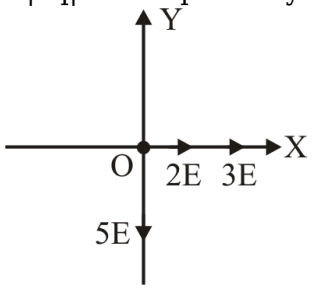
1) **Explaining :** Find the ratio of electric field of the sphere.
Concept : This question is based on relation between electric field & potential.
Solution : $V_1 = V_2$

$$\left(\frac{\sigma_1}{\varepsilon_0}\right) a = \left(\frac{\sigma_2}{\varepsilon_0}\right) b$$

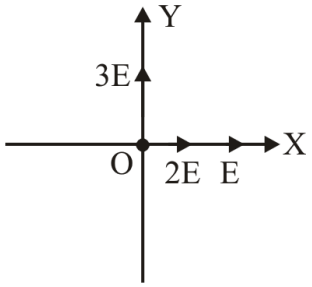
$$\frac{E_1}{E_2} = \frac{b}{a}$$

Final Answer : (2)

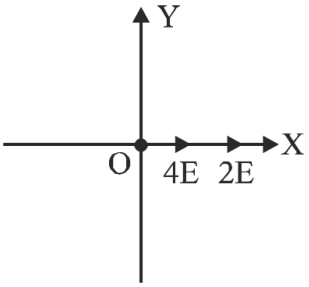
2) **Explaining :** Find net electric field at origin.
Concept : This question is based on superposition of electric field.
Solution :
If electric field due to charge |q| at origin is E then electric field due to charges |2q|, |3q|, |4q| and |5q| are respectively 2E, 3E, 4E and 5E



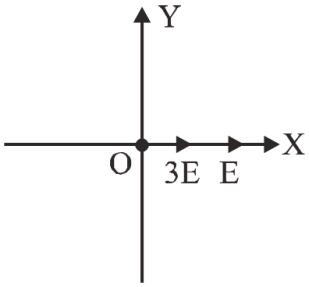
(i)



(ii)



(iii)



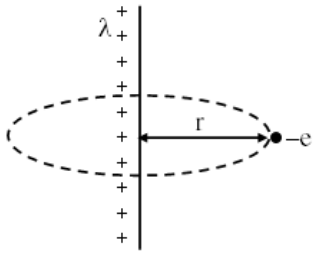
(iv)

$$E_{(i)} = \sqrt{(5E)^2 + (5E)^2} = 5\sqrt{2}E,$$

$$E_{(ii)} = \sqrt{(3E)^2 + (3E)^2} = 3\sqrt{2}E,$$

E(iii) = 4E + 2E = 6E and E(iv) = 3E + E = 4E
$\Rightarrow E_{(i)} > E_{(iii)} > E_{(ii)} > E_{(iv)}$
Final Answer : (3)

3) **Explaining :** An infinitely long positively charged straight thread has a linear charge density λ Cm^{-1}. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :
Concept : This question is based on electrostatic force provide necessary centripetal force.
Solution :



Electric field E at a distance r due to infinite long wire is $E = \frac{2k\lambda}{r}$

Force of electron ⇒ F = eE

$$F = e\left(\frac{2k\lambda}{r}\right)$$

$$F = \frac{2k\lambda e}{r}$$

This force will provide required centripetal force

$$F = \frac{mv^2}{r} = \frac{2k\lambda e}{r}$$

$$v = \sqrt{\frac{2k\lambda e}{m}}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{2k\lambda e}{m}\right) = k\lambda e$$

This is constant so option (2) is correct.

Final Answer : (2)

4) **Explaining :** Find electric flux through the cube.

Concept : This question is based on electric flux.

Solution : $\phi_1 = -E_1a^2 = -\alpha a^{1/2}a^2 = -\alpha a^{5/2}$ and $\phi_2 = E_2a^2 = \alpha(2a)^{1/2}a^2 = \sqrt{2}\alpha a^{5/2}$

$\phi_{net} = (\sqrt{2} - 1)\,\alpha a^{5/2}$

Final Answer : (3)

5) **Explaining :** Electric field of ring, shell, solid sphere.

Concept : This question is based on grap.

Solution : Theoritical.

Final Answer : (1)

6) **Explaining :** $E = -\frac{dv}{dr}$

Concept : This question is based on relation between electric field and potential.

Solution : $\vec{E} = (-4x + 3y^2z)\hat{i} + (6xyz - 4z)\hat{j} + (3xy^2 - 4y)\hat{k}$

at (0, 1, 1) → $\vec{E}_p = 3\hat{i} - 4\hat{j} - 4\hat{k}$

Final Answer : (3)

7) **Explaining :** Potential energy.

Concept : This question is based on potential energy.

Solution : Net electrostatic energy of the configuration will be :

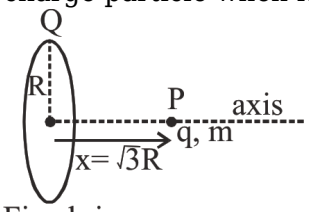
$$U = K\left[\frac{q \cdot q}{a} + \frac{Q.q}{\sqrt{2a}} + \frac{Q.q}{a}\right]$$

Here, $k = \frac{1}{4\pi\varepsilon_0}$

U = 0 gives, $Q = \frac{-2q}{2+\sqrt{2}}$

Final Answer : (2)

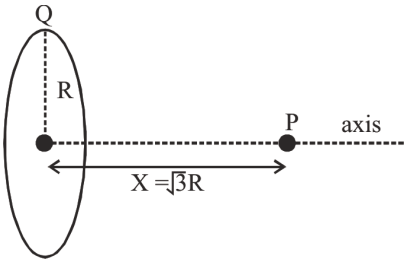
8) **Explaining :** A point charge 'q' of mass is located at the axis of a ring at point P($x = \sqrt{3}R$) oriented along x-axis. When the point charge q is given negligible velocity. The speed of charge particle when it reaches to infinity is :-



Fixed ring

Concept : This question is based on conservation energy.

Solution :



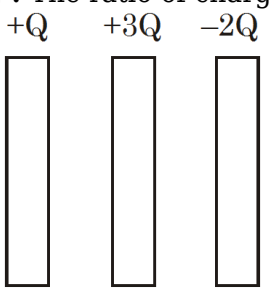
Energy conservation between point 'P' and ∞

$$0 + q\frac{KQ}{\sqrt{R^2+3R^2}} = \frac{1}{2}mv^2 + 0$$

$$\frac{KQq}{2R} = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{KQq}{mR}}$$

Final Answer : (1)

9) **Explaining :** The ratio of charge on left surface of plate A to the charge on the right surface of plate B is :-



Concept : This question is based on conservation of charge.

Solution :

The charge on the outer surface's of plate A and plate C will be = $\frac{Q+3Q-2Q}{2} = Q.$

So the charge on left surface of plate 'A' = Q

$\because$ Charge on right surface of 'A' will be zero on left surface of plate 'B' would be zero.

$\therefore$ Charge on right surface of 'B' will be 3Q

Final Answer : (2)

10) **Explaining :** $E = -\frac{dv}{dr}$

Concept : This question is based on relation between electric field & potential.

Solution : $E_x = -\frac{dv}{dx}$, $E_x = -\left(\frac{2}{2 \times 10^{-2}}\right)$ = – 100 N/C,

$E_y = -\left(\frac{-2}{1 \times 10^{-2}}\right)$ = 200 N/C

Final Answer : (2)

11) **Explaining :** dv = –Edr.

Concept : This question is based on dv = –Edr.

Solution : $dV = -\vec{E}.\vec{dr}$

$Vp - V_o = -\vec{E}.(\vec{r_p} - \vec{r_o})$

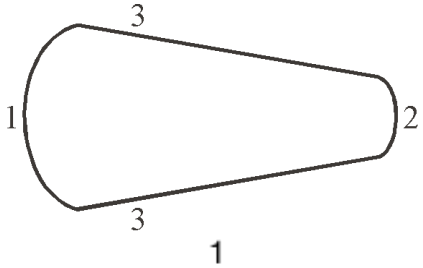
$V_p - 150 = -(20\hat{i} + 30\hat{j})(2\hat{i} + 2\hat{j})$

$V_p - 150 = -(40 + 60) = -100$

$V_p = 50$ V

Final Answer : (3)

12) **Explaining :** Consider a non-spherical conductor shown in the figure which is given a certain amount of positive charge. The charge distributes itself on the surface such that the charge densities are σ_1, σ_2 and σ_3 at the regions 1, 2 and 3 respectively. Then :-

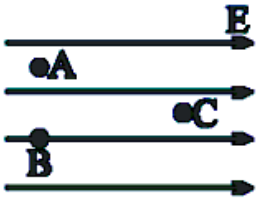


Concept : This question is based on $\sigma \propto \frac{1}{r}$.

Solution : $\sigma \propto \frac{1}{r}$

Final Answer : (4)

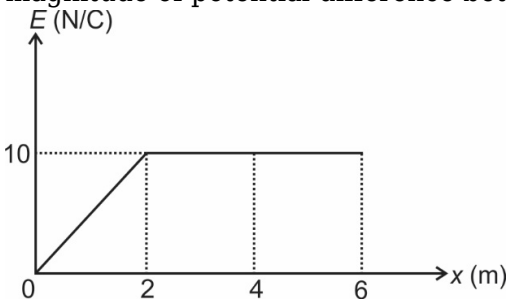
13) **Explaining :** Suppose a region of space has a uniform electric field, directed towards the



right, as shown below. Which statement is true? **Concept :** This question is based on equipotential surface.
Solution : Direction of electric field is always from HP to LP.
Final Answer : (3)

14) **Explaining :** There is a uniform electric field in X-direction. If the work done by external agent in moving a charge of 0.2 C through a distance of 2 metre slowly along the line making an angle of 60° with X-direction is 4 joule, then the magnitude of E is :
Concept : This question is based on W = Fr cos θ.
Solution : W = Fr cos θ ⇒ 4 = (0.2) E (2) cos 60°
⇒ E = 20 N/C.
Final Answer : (4)

15) **Explaining :** Figure shows the variation of electric field intensity E versus distance x. The magnitude of potential difference between the points x = 0 to x = 5 m is



Concept : This question is based on area under the curve b/w E–x gives potential difference.
Solution : Potential difference = area under E – x graph
$= \frac{1}{2} \times 2 \times (10) + (10)(5 - 2) = 10 + 30 = 40$ V
Final Answer : (3)

16) **Explaining :** A $500\mu F$ capacitor is charged at a steady rate of 100μC/ second. The potential difference across the capacitor will be 10V after an interval of :-
Concept : This question is based on q = CV.
Solution : q = CV

$$\frac{dq}{dt} = C\frac{dV}{dt}$$

$$\Delta t = \frac{C\Delta V}{\frac{\Delta q}{\Delta t}} = \frac{(500 \times 10^{-6})(10)}{100 \times 10^{-6}} = 50\,\text{sec}$$

Final Answer : (4)

17) **Explaining :** The radius of the circular plates of a parallel plate capacitor is 'R'. Air is dielectric medium between the plates. If the capacitance of the capacitor is equal to the capacitance of sphere of radius 'R', then the distance between the plates is :

Concept : This question is based on $C = \frac{\varepsilon_0 A}{d}$ & $C = 4\pi\varepsilon_0 R$.

Solution : $\frac{\varepsilon_0 A}{d} = 4\pi\varepsilon_0 R$

 plates are circular $\frac{\varepsilon_0 \pi R^2}{d} = 4\pi\varepsilon_0 R$

$d = \frac{R}{4}$

Final Answer : (1)

18) **Explaining :** A 40 mF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is :-

Concept : This question is based on $P = \frac{W}{t}$.

Solution : $P = \frac{W}{t} = \frac{\frac{1}{2}CV^2}{t}$

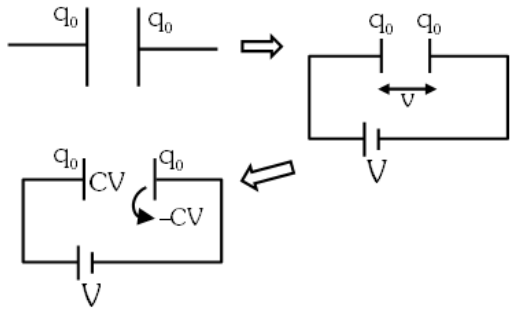
$P = \frac{40 \times 10^{-3} \times 9 \times 10^6}{2 \times 2 \times 10^{-3}} = 90MW$

Final Answer : (2)

19) **Explaining :** A parallel plate capacitor of capacitance C is having charge q_0 on each plate. Now it is connected to a battery of emf V then which option is correct ?
(i) Charge supplies by battery would be CV
(ii) The outer surface of plate have equal charge q_0
(iii) The facing surface have equal and opposite charge of magnitude CV

Concept : This question is based on Q = CV.

Solution :



Final Answer : (4)

20) **Explaining :** The plates of a parallel plate capacitor are pulled apart with a velocity v. If at any instant their mutual distance of separation is x then magnitude of rate of change of capacitance with respect to time varies as :

Concept : This question is based on $C = \frac{\varepsilon_0 A}{x}$.

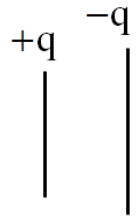
Solution : $C = \frac{A\varepsilon_0}{x}$

$\frac{dc}{dt} = -\frac{A\varepsilon_0}{x^2}\frac{dx}{dt} = \frac{A\varepsilon_0 v}{x^2}$ showing

$\frac{dc}{dt} \propto x^{-2}$

Final Answer : (2)

21) **Explaining :** If the plates of a parallel plate capacitor are not equal in area, then:
Concept : This question is based on induction effect.
Solution : Induction effect



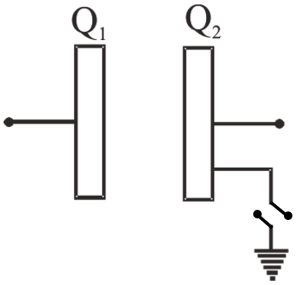
Final Answer : (1)

22) **Explaining :** A battery is used to charged a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will doubled the :
Concept : This question is based on d' = 2d.
Solution : d' = 2d ,c' = c/2, q' = q,

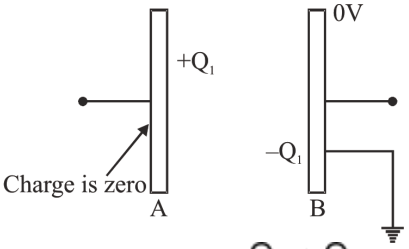
$U' = \frac{q'^2}{2c^1} = \frac{q^2}{2c/2} = 2U$

Final Answer : (3)

23) **Explaining :** The two plates of a parallel plate capacitors charges Q_1 and Q_2 are given.The capacity of the capacitor is C. When the switch is closed, mark the correct



statement(s).Assuming both Q_1,Q_2 to be positive. **Concept :** This question is based on earthing.
Solution : When we earth plate B, whole of its charge will go to earth. Now charge of plate A will induce charge –Q and $+Q_1$ on inner side and outer side of plate B. Again Q_1 charge of plate B will go to earth. $-Q_1$ charge will not go to earth. It will be bounded to +Q, charge of plate A



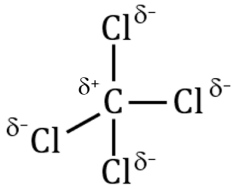
So, it is clear that $\frac{Q_1 + Q_2}{2}$ charge goes from 2nd plate to earth.

Charge of capacitor is Q_1 and hence its potential is $\frac{Q_1}{C}$.

Final Answer : (3)

CHEMISTRY

24) (A) The process of adding an e^- to a neutral gaseous atom is not always exothermic it may be exothermic or endothermic.
(B) Ionisation energy is always positive.
(C) Be $1s^2 2s^2$ B $1s^2 2s^2 2p^1$
In Be 2s subshell in fully filled
So, need high energy to remove e^- as compared to B.

(D) In $CCl_4 \rightarrow$ 

due to partially positive charge $Z_{eff} \uparrow$, EN$\uparrow$
So, EN of C ⇒ $CCl_4 > CH_4$
(E) Cs is most electropositive
Final Answer : (2)

25) The metallic radius order of Al & Ga is

B < Ga < Al < In < Tl

(due to poor shielding of d-subshell electrons)
The ionic radius order of Al^{+3} & Ga^{+3} is
$B^{+3} < Al^{+3} < Ga^{+3} < In^{+3} < T\square^{+3}$
Final Answer : (3)

26) Higher difference between ionisation enthalpy and election gain enthalpy is for compound EB. So, EB has higher ionic character.
The Lower value of difference between ionisation enthalpy and election gain enthalpy is for compound ED. So, ED has lower ionic character. The increasing order of compounds for the difference between ionisation enthalpy and electron gain enthalpy →
EB < EC < EA < EB
So, increasing. order of compounds for ionic character.
ED < EC < EA < EB
In the decreasing order,
EB > EA > EC > ED
Final Answer : (1)

27) (1) In a period, from left to right the acidic strength increases that mean basic nature decreases.
(2) In a group, from top to bottom the basic nature increases that means acidic nature decreases.

Final Answer : (2)

28) (i) I shows only one O.S. i.e. +1
(ii) II is f-block element
(iii) The compound formed between I and III is ionic.
(iv) IV is d-block element and shows variable O.S.
Final Answer : (4)

29)

HF forms strong Hydrogen bonding with other HF molecules but HCl, HBr, HI do not form H-bonding.

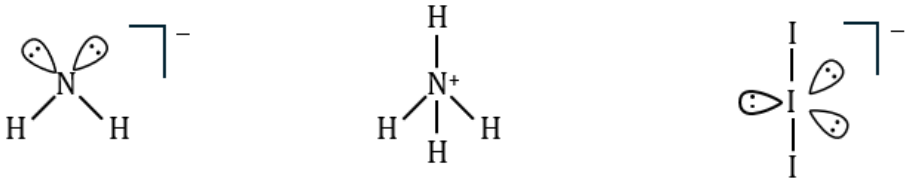
30) (i) $O \xrightarrow{\Delta H_1} O^- \xrightarrow{\Delta H_2} O^{2-}$
(ii) $S \xrightarrow{\Delta H_3} S^- \xrightarrow{\Delta H_4} S^{2-}$
(iii) $P \xrightarrow{\Delta H_5} P^- \xrightarrow{\Delta H_6} P^{2-}$
(a) ΔH_5 is less negative than ΔH_3 and ΔH_1 due to half field stable configuration.
(b) ΔH_1 is less negative than ΔH_3 due to more electron-electron repulsion in O-atom then in S-atom.
(c) ΔH_2 is less negative than ΔH_4 due to more electron-electron repulsion in O^--ion then in S-ion.
Final Answer : (4)

31) internuclear axis - z-axis
$d_{yz} + d_{z^2}$ - no bonding
$d_{yz} + d_{xy}$ - no bonding
$d_{x^2-y^2} + d_{xy}$ - no bonding
Final Answer : (4)

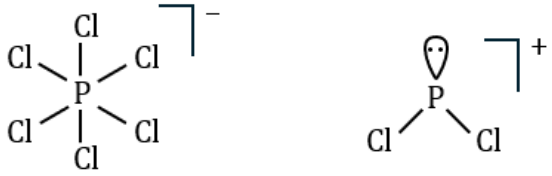
32) **Explaining :** Select the incorrect statement:
Concept : This question is based on hybridization.
Solution : PF_5 and PCl_5 exist through the formation of sp^3d-hybridization.
Final Answer : (4)

33) Xenon-H_2O → Dipole-induced dipole interaction
Ethanol-H_2O mixture → Dipole-dipole interaction
Liquid He → London forces
Liquid HCl → Dipole-dipole interaction
Final Answer : (2)

(a) NH_2^- (b) NH_4^+ (c) I_3^-



(d) PCl_6^- (e) PCl_2^+



34)

Final Answer : (3)

35) $PI_{5(g)}$

(1) Due to the size of iodine, which is quite enormous, five iodine atoms cannot fit around the P atom to form phosphorus pentaiodide.

$PI_{5(S)} \rightarrow PI_4^+ + I^-$

Pl_5 appears to be $[Pl_4]^+$ and I^- in solution.

(2) All possible angle in BF_2CI is not equal to 120 degree.

(3) Since Si possesses empty d-orbitals, $N(SiH_3)_3$ has a planar structure because the lone pair N-atom is moved to the silicon's empty d-orbitals (pπ-dπ overlap).

(4) $PBr_{5(s)} \rightarrow PBr_4^+ (sp^3) + Br^-$

d-orbital is not used in hybridisation

All the options are false. Hence, the option (3) is correct.

Final Answer : (3)

36) Molecules with non-zero dipole moment

SO_2 (bent)

NF_3 (pyramidal)

NH_3 (pyramidal)

ClF_3 (T-shaped)

SF_4 (seesaw)

(XeF_2 is linear → zero dipole)

Lone-pair count on central atom

SO_2: 1

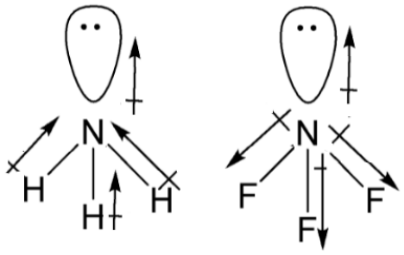
NF_3: 1

NH_3: 1

SF_4: 1

ClF_3: 2

Final Answer : (4)

37) 

Dipole moment of NH_3 is higher than NF_3 as μ of lone-pair is in same direction as the resultant dipole moment of N—H bonds.
So, both (A) & (R) are true, & (R) is the correct explanation of (A)
Final Answer : (4)

38)

Species	**B.O.**
O_2^{2-}	1
CO	3
$NO^{\oplus}$	3

Final Answer : (1)

39) p = 3 q = 5 r = 2
s = 6, t = 2
3 + 5 + 2 – 6 – 2 ⇒ 2
Final Answer : (1)

40)

Due to inter molecule H-bonding p-nitrophenol has more boiling point than o-nitrophenol. Hence o-nitrophenol is more volatile than p-nitrophenol.
Final Answer : (3)

41) **Solution :**
Volume of ice > volume of water and thus, increase in pressure favours forward reaction showing decrease in volume.
Final Answer : (2)

42) **Solution :**
Addition of reactants shifts the reaction from left to right, i.e., favours forward reaction.
Final Answer : (1)

43) **Solution :**
$NH_4Cl_{(s)} \rightleftharpoons NH_{3(g)} + HCl_{(g)}$
$\Delta n = 2 - 0 = 2$
Final Answer : (4)

44) **Solution :**

$$\underset{\text{At eq.}}{PCl_{5(g)}} \rightleftharpoons \underset{0.8 \times 10^{-3}}{PCl_3} + \underset{1.2 \times 10^{-3}}{Cl_2} \quad \underset{1.2 \times 10^{-3}}{}$$

$$K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{1.2 \times 10^{-3} \times 1.2 \times 10^{-3}}{0.8 \times 10^{-3}} = 1.8 \times 10^{-3}$$

Final Answer : (2)

45) **Solution :** Le Chatelier's principle is not valid for solid-solid equilibrium.
Final Answer : (3)

BIOLOGY

46)

Explain Question : The question is asking about bacteria that oxidize inorganic substances (like ammonia, nitrites, H_2S, Fe^{2+}) and use the released energy to produce ATP.

Concept : This question is based on Nutrition in monera

Solution Bacteria that oxidize inorganic substances (like ammonia, nitrites, H_2S, Fe^{2+}) and use the released energy to produce ATP. Such bacteria are called Chemosynthetic autotrophs. Examples: Nitrosomonas and Nitrobacter
Final answer_3

47) **Explain Question :** The question is asking about group of organisms plays the most important role in recycling nutrients (nitrogen, phosphorus, iron, sulfur) in nature
Concept : This question is based on Monera
Solution : Monera (bacteria) includes chemosynthetic autotrophic bacteria which recycle nutrients like N, P, Fe, S in nature.
Final answer : (1)

48)

Question explanation: Question asking about which are basis of modern taxonomic studies

Concept: Taxonomy type, definition

Solution: In modern taxonomical studies, taxonomists use external and internal structure, along with the structure of cell, developmental processes and ecological information of organisms

Final answer: option (3)

49)

Explain Question: Question asking about the earliest classifications were based on which characters

Concept: Type of classification

Solution:
Before scientific systems like those of Aristotle or Linnaeus, the very earliest classification attempts by ancient humans were indeed based on the uses of organisms — for example:
Plants classified as useful (food, medicine, fiber) or harmful (poisonous, weeds).
Animals grouped as domestic, wild, edible, etc.

Final answer: 1

50)

Question Explanation: The question asks which statements about biological names, taxa, and plant families are correct.

Concept: The Living World → Binomial nomenclature & Taxonomic categories.

Solution:
(a) Correct – Biological names are written in Latin, and italicized when printed.
(b) Correct – Each taxon is a unit of classification in the taxonomic hierarchy.
(c) Correct – *Datura* and *Petunia* both belong to the family Solanaceae.
All statements are correct.

Final answer: (2)

51) **Solution :** A-Dicotyledonae, B-Sapindales, C-Anacardiaceae
Final Answer : (3)

52) **Solution :**
(b) Mammalia (Class)
(d) Carnivora (Order) (c) *Felidae* (Family)
(a) *Panthera* (Genus) **Final Answer : (4)**

53)

Explain Question : The bacterial cell is very simple in A but very complex in B. You are to select the correct pair (A and B) that completes this statement accurately

Concept : This question is based on Kingdom Monera

Solution : The correct answer is 4. A = Structure, B = Behaviour
Explanation: Bacterial structure is very simple, they are very complex in behaviour Bacteria are structurally simple:
–No nucleus or membrane-bound organelles
Prokaryotic cell structure
So, A = Structure

Despite their simplicity, bacteria show complex behavior:
– Adaptation, resistance, gene exchange (conjugation, transformation)
So, B = Behaviour

Final Answer : (4)

54) **Explain Question:** Matching protozoan groups with their correct examples/features.
Concept: Protista
Solution: Correct match is -
Ciliated protozoans → *Paramecium (Paramoecium).*
Sporozoans → *Plasmodium* (malaria parasite).
Flagellated protozoans → cause sleeping sickness (*Trypanosoma*).
Amoeboid protozoans → move by pseudopodia. (P–iii, Q–iv, R–ii, S–i.) **Final answer:** (4)

55)

Explain Question : Identify which genus–character–class combo is incorrectly matched.
Concept : This question is based on Fungal classes
Explanation :Deuteromycetes: no sexual stage (e.g., Alternaria).
Ascomycetes: endogenous ascospores (e.g., Aspergillus).
"Phycomycetes" (incl. zygomycetes/oomycetes): mostly aseptate/coenocytic hyphae (e.g., Mucor).
Albugo is an oomycete with aseptate coenocytic hyphae, not Ascomycete.
Final answer : 4

56)

Explain Question: The question is asking about how much time it will take to fill 1/4 of the cup ? If a bacteria divides once every 3 minutes and fills the cup in 57 minutes.
Concept : This question is based on Binary fission
Solution : Bacteria divide once every 3 minutes and the cup is full in 57 minutes.That means at 54 minutes the cup was half full (because in the next 3 minutes it doubles to fill completely). At 51 minutes the cup was ¼ full (because in the next 3 minutes it doubles to ½ full, and then in another 3 minutes it becomes full).
Final answer : 2

57) **Explain Question :** The question asks which organism causes the phenomenon called red tide.
Concept : This question is based on Dinoflagellates
Explanation : Red tide → Gonyaulax.
Final answer : (3)

58) **Explain Question:** Question is asking about identifying the incorrect pair between bacterial structures/features and their
functions.
Concept: This question is based on Structure of Bacterial cell
Solution: All pairs are correct except "Pili – Locomotory structure," since pili help in gene transfer/adhesion, not movement.

Final answer: (4)

59) **Explain Question:** Question is asking about how many of the given organisms belong to **sac fungi (Ascomycota)**. **Concept:** This question is based on Ascomycetes
Solution: Members of sac fungi = Penicillium, Aspergillus, Claviceps, Morels, Neurospora, Yeast → **6**.
Final answer: (2)

60) **Solution :** Cyanobacteria are eubacteria, they do not have chloroplast.
Final Answer : (4)

61)

Explain Question: Question asks to identify the correct matches of bacterial structures (Column I) with their properties/functions (Column II & III).
Concept: Structure of Bacteria
Solution: Mesosomes: infoldings of plasma membrane, contain oxidative enzymes, help in respiration → correct.
Capsule: loose sheath around cell, but not for protein synthesis (function is protection) → incorrect.
Flagella: made of flagellin protein, used for motility, not for genetic recombination → incorrect.
Final answer: (4)

62) **Explanation:**

A. Statement I:
B. Kingdom Protista includes diverse unicellular eukaryotic organisms like *Chlamydomonas*, *Chlorella* (photosynthetic algae), *Paramecium* (ciliate), and *Amoeba* (amoeboid protozoan).
C. In earlier classifications, photosynthetic algae (*Chlamydomonas* and *Chlorella*) were placed in the Plant Kingdom, while heterotrophic organisms like *Paramecium* and *Amoeba* were classified under the.

Animal Kingdom. Protista brought these together based on their unicellular eukaryotic nature.

A. Therefore, Statement I is correct.
B. Statement II:
C. The classification criteria evolved to emphasize cellular structure (prokaryotic vs. eukaryotic) and level of organization (unicellular vs. multicellular), rather than just nutrition or habitat.
D. This led to the reorganization of unicellular eukaryotes into a single kingdom, Protista.
E. Hence, Statement II is also correct.

Both statements are accurate and complement each other.
Final Answer : (4)

63)

Explain Question : The question asks about the composition of a pyrenoid, a structure found within the chloroplasts of many algae.

Concept : This question is based on Chlorophyceae.

Solution : A pyrenoid, characteristic of algae, primarily consists of a core of protein surrounded by starch.

Final Answer : (2)

64) **Solution :**

	List-I (Type of Sexual Reproduction)		**List-II (Examples)**
(a)	Filamentous Algae of Chlorophyceae	(iV)	*Ulothrix*
(b)	Filamentous Algae of Phaecophyceae	(iiI)	*Ectocarpous*
(c)	profusely branched forms of Phaeophyceae	(i)	*Kelps*
(d)	Colonial Algae of Chlorophyceae	(iI)	*Volvox*

Final Answer : (2)

65) **Explaining the Question:** The question asks which fungi are responsible for causing rust and smut disease in plants.
Concept: Classification of fungi based disease.
Solution:
Rust disease - Caused by fungi of the genus *Puccinia*.
Smut disease - Caused by fungi of genus *Ustilago*.
Final Answer: (2)

66) **Explain Question:** Question asks about which protistan group has a protein-rich flexible layer around the body.
Concept: Euglenoids
Solution: Organisms with a protein-rich flexible covering → Euglenoids.
Final answer: (4)

67)

Explain Question: Question asks which statements correctly describe the organism shown in the diagram (a dinoflagellate).
Concept: Dinoflagellate
Solution:
(a) Incorrect – Dinoflagellates show many colors but not green.
(b) Correct – Some species release toxins.
(c) Correct – Two flagella differ in position.
(d) Incorrect – Wall made of cellulose plates, not peptidoglycan.
Final answer: (2)

68) **Solution :** *Neurospora* is members of class ascomycetes and it is used extensively in genetic work, so also known as "Drosophila of plant kingdom".
Final Answer : (1)

69) **Question Explanation:** Analyzing an Assertion-Reason pair about the common name and defining characteristic of Deuteromycetes.
Concept: Defining features of Fungi Imperfecti.
Solution: The **Assertion** is **True**. **Deuteromycetes** are correctly called **Imperfect Fungi**. The **Reason** is also **True**. The classification is based on the fact that only the **asexual or vegetative phases** of these fungi are known, and the sexual (or "perfect") phase is missing or yet to be discovered. Since the Reason (R) provides the exact structural and reproductive justification for the name given in the Assertion (A), the Reason is the **correct explanation** of the Assertion.**Final Answer:** Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
Final Answer : (1)

70)

Explain Question : The question is asking which statements about **methanogens** are correct.
Concept : This question is based on Archaebacteria.
Solution :
(i) Correct → Archaebacteria.
(ii) Correct → biogas production.
(iii) Incorrect → thermoacidophiles live in hot sulphur springs.
(iv) Correct → anaerobic.
Final answer : (2)

71)

Explain Question:
Question asking about life cycle of fern (pteridophyta)
Concept: This question is based on Pteridophyta -general characters and life cycle
Solution:
In fern the spores germinate to give rise to inconspicuous, small but multicellular free-living, mostly photosynthetic thalloid gametophytes called prothallus.
Final answer: (4)

72)

Question Explanation: Question asks about matching plant structures/groups in Column A with their correct associations in Column B.
Concept: Pteridophytes and Gymnosperms
Solution:
Mycorrhiza → found in *Pinus*
Coralloid roots → found in *Cycas*
Male gametophyte → Pollen grain
Selaginella → belongs to Lycopsida
Giant redwood → *Sequoia*
Final Answer: (3)

73) **Solution :** Mosses are attached to soil through multicellular and branched rhizoids.

Final Answer : (3)

74) **Question Explanation:** The question asks to identify which is not typically considered an edible algae.
Concept : The question is based on algae - Phaeophyceae
Solution: Fucus is a brown algae primarily used for its medicinal properties (e.g., iodine source) and as a supplement, but it is not widely consumed as a food source
Final answer : (4)

75) **Question Explanation :** The question asks to find the organism that does not belong to the same major taxonomic group
Concept : The question is based on class of algae
Solution : *Polysiphonia*, *Porphyra*, and *Gelidium*—are members of the class Rhodophyceae (red algae). Sargassum belongs to the class *Phaeophyceae* (brown algae)
Final answer : (3)

76) **Solution :**

Column-I		**Column-II**	
A	*Cycas*	iii	Pinnate leaves
B	*Pinus*	iv	Needle like leaves
C	*Salvinia*	i	Heterosporous fern
D	*Adiantum*	ii	Homosporous fern

Final Answer : (3)

77)

Explain Question : Question asking select correct statement regarding gymnosperm.
Concept : This question is based on Gymnosperm-general characters ,life cycle.
Solution :
1 Seeds are developed pre-fertilization-Incorrect
Correct is-The seeds that develop post-fertilisation,
2 Ovules are not enclosed by any ovary wall and remain exposed, both before and after fertilization-correct
3 They show double fertilization--Incorrect
Correct is-double fertilisation characteristic feature of angiosperm
4 They form haploid endosperm after fertilization--Incorrect
In gymnosperm endosperm form before fertilization and it is a haploid endosperm.
Final answer : 2

78) **SOLUTION-**
In pteridophytes, the prothallus produces antheridia and archegonia.
Explanation:
• The prothallus is the gametophyte stage in the life cycle of pteridophytes.
• It produces gametes: antheridia (male gametes) and archegonia (female gametes).

• These structures are essential for sexual reproduction in ferns and other pteridophytes.
Correct answer: Antheridia and archegonia.
Final Answer : (2)

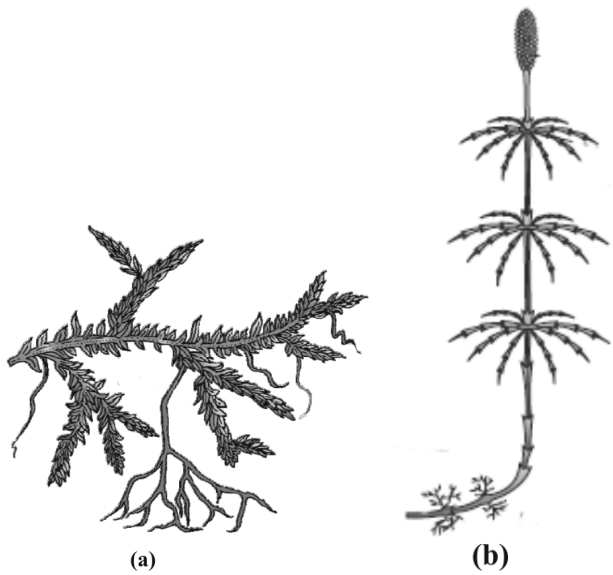
79)

Explain Question : Question asking select incorrect statement about life cycle of pteridophytes
Concept : This question is based on pteridophytes
Solution :
1 Main plant body of pteridophytes is differentiated into root, stem and leaves.-**correct**
2 Large and small types of leaves are present in pteridophytes -**correct**
3 Pteridophytes are the first terrestrial plants to possess xylem and phloem.-**correct**
4 Most of the pteridophytes produce two kinds of spores, macro and micro spores by meiotic division. **Incorrect**
In the majority of the pteridophytes all the spores are of similar kinds; such plants are called homosporous. Some pteridophytes (Genera like Selaginella and Salvinia) which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous.

Final Answer : (4)

80)

Explain Question : Question asking about Identifying the plants given in the figure (a) and (b).



Concept : Intra of Pteridophytes
Solution : The given figures (a) is *Selaginella* and (b) is *Equisetum*
Final answer : (3)

81) **Explain Question :**The question asks to identify which of the provided statements are not considered an ecological importance of mosses.

Concept : This question is based on mosses.
Solution :C is not an ecological importance but rather a human use of mosses.
Final answer : 3

82) **Question Explanation :** The question provides two statements about bryophytes and asks you to evaluate whether they are true or false, and whether Statement-2 correctly explains Statement-1.
Concept: This question is based on Bryophytes.
Solution: Statement-1: Bryophytes are also called amphibians of the plant kingdom **- True**
Statement-2: Bryophytes can live in soil but are dependent on water for fertilization **- True**
Bryophytes are also called amphibians of the plant kingdom because these plants can live in soil but are dependent on water for sexual reproduction. They usually occur in damp, humid and shaded localities. **Final answer: (4)**

83) **Explain Question :**The question has provided an image showing three figures of bryophytes, with different parts labeled as P, Q, R, and S. The task is to correctly identify these labeled structures.
Concept : This question is based on bryophytes (mosses and liverworts).
Explanation :
P : Archegoniophore
Q : Gemma cup
R : Antheridiophore
S : Capsule.
Final answer : 4

84) **Explain Question :**The question asks to identify the correct description for the stage labeled "X" in flowchart of the life cycle of a moss
Concept : This question is based on life cycle of a moss
Explanation :"Creeping green, branched and filamentous stage" This is the precise description of a "protonema"
Final answer : (4)

85) **Explain Question :** The question provides a list of algae genera and asks How many of them belong to the class Chlorophyceae (green algae)
Concept : This question is based on Chlorophyceae (green algae)
Explanation :
"Chlorophyceae (green algae) are mostly freshwater and have chlorophyll a and b as major pigments. Common members include Volvox, Chlorella, Ulothrix, and Chara."
Final answer : (2)

86) **Explain Question :** The question asks you to identify a specific type of alga that is known for its high protein content and has been used as a food supplement, particularly by space travelers
Concept : This question is based on Chlorophyceae
Explanation : "Chlorella is a unicellular green alga rich in protein and used as a food supplement by space travellers."

Final answer : (2)

87) **Explain Question :**The question asks you to analyze an Assertion and a Reason and determine the correct relationship between them
Concept : This question is based on Phaeophyceae
Solution :
Assertion:Fucus has large, non-motile female gametes and small, motile male gametes.
True — matches the definition of oogamy.
Reason:Fucus shows oogamous reproduction.
True — Correct classification of the reproductive strategy.
Oogamy directly explains the fusion of a large, non-motile egg with a small, motile sperm.
Final answer : (1)

88) **Explain Question :**The question asks to identify the specific group of algae based on the unique characteristics of their gametes.
Concept : This question is based on Brown algae (Phaeophyceae)
Explanation : The gametes of brown algae are pyriform and have two laterally inserted flagella of unequal length.
Final answer : (3)

89) **Correct Answer:** Option 1: II, III, I, IV
Explanation: The life cycle of a fern (pteridophyte) follows the haplodiplontic pattern, alternating between the sporophyte (2n) and gametophyte (n) stages. The correct sequence of events is:

A. Liberation of Spores (II) – The sporophyte (diploid) produces spores (haploid) through meiosis, which are released into the environment.
B. Prothallus Formation (III) – The haploid spores germinate to form a small, independent gametophyte (prothallus). This prothallus carries out photosynthesis and develops reproductive structures.
C. Fertilization (I) – The antheridia (male) release motile sperm, which swim in water to fertilize the egg in the archegonia (female), forming a diploid zygote.
D. Embryo Formation (IV) – The zygote develops into a young sporophyte (embryo), which later matures into the dominant fern plant.

Final Answer : (1)

90)

Question Explanation: The question is asking which type of algae stores laminarin and mannitol as their food reserves.
Concept: Algae - Phaeophyceae
Solution: In brown algae food is stored as complex carbohydrates, which is the form of laminarin or mannitol.
Final answer: (2)